Practical Optimization —Linear Programming

Exercises

EXERCISE 1

Consider the following linear program:

$$\begin{aligned} \text{maximize} \quad & x_1 - x_2 + x_3, \\ \text{subject to} \quad & 4x_1 + 3x_2 - x_3 = 1, \\ & 6x_1 - 2x_2 + 4x_3 \geq 8, \\ & x_1 - x_2 \leq 5, \\ & x_1 \geq 0, \quad x_2 \leq 0. \end{aligned}$$

(1) Formulate the dual problem.

(2) Check that $x^* = (5, 0, 19)$ is feasible for the primal and that $y^* = (-1, 0, 5)$ is feasible for the dual.

(3) Without solving the problem, explain why the pair (x^*, y^*) is a primal–dual optimal solution.

EXERCISE 2 (Chebyshev approximation)

We consider the set of data in the file `cheb.dat` (on the Community platform). The goal is to approximate this set of points by a polynomial of degree $n = 5$ by solving different optimization formulations.

Let $P(x) = \sum_{i=0}^{n} p_i x^i$.

(1) The Chebyshev approximation problem is

$$\min_{p \in \mathbb{R}^{n+1}} \max_{k \in \{1,\dots,m\}} \left| P(x_k) - y_k \right|.$$

Reformulate this as an LP (introduce the slack variable for the maximum absolute error). Modelize the LP in AMPL and solve it with SNOPT.

(2) Let $Q(x) = \sum_{i=0}^{n} q_i x^i$. Compute the coefficients q by solving the linear least-squares problem

$$\min_{q \in \mathbb{R}^{n+1}} \sum_{k=1}^{m} \big(Q(x_k) - y_k\big)^2.$$

(Standard normal equations or QR/LS solver.)

(3) Solve the ℓ_1-approximation problem. Let $R(x) = \sum_{i=0}^{n} r_i x^i$. Introduce variables $u_k \geq 0$ and $v_k \geq 0$ for $k = 1, \ldots, m$ and constraints

$$R(x_k) - y_k = u_k - v_k, \qquad k = 1, \ldots, m,$$

and minimize $\sum_{k=1}^{m}(u_k + v_k)$. What is the class of this problem? Solve it to find R.

(4) Compare the three solutions by computing the residual norms $\|\cdot\|_2, \|\cdot\|_1, \|\cdot\|_\infty$. Present a 3×3 table with norms in the columns and polynomials P, Q, R in the rows. Plot the three polynomials and the data points on the same plot and comment on the results.

EXERCISE 3 (Textiles firm)

A textiles firm buys pants from manufacturers and sells them to clothing retailers. There are four styles with total orders (in thousands of pants, i.e. units = K). Three manufacturers M_1, M_2, M_3 can make these styles in quantities and prices (/pant) as given below.

	M1		M2		M3	
Style	Capacity (K)	Price ()	Capacity (K)	Price ()	Capacity (K)	Price ()
1	200	8	80	6.75	120	9
2	150	10	60	10.25	100	10.50
3	90	11	50	11	75	10.75
4	70	13	40	14	50	12.75
Total capacity	275		150		220	

Manufacturing facility M_3 is domestic (no import limit). Manufacturers M_1 and M_2 are foreign and together are limited by an import quota of 350K pants. Ignore integer constraints (continuous variables).

(1) Formulate the problem as an LP.

(2) Write an AMPL model and solve using MINOS, then LOQO.

(3) Compare the two solvers: optimal value, optimal solution, number of iterations explain differences (if any).

(4) Is it possible that the number of pants of style 3 bought from manufacturers 1 and 2 are equal while staying optimal? Explain.

(5) If the quota increases from 350K to 355K, estimate the variation of optimal cost using the relevant dual variable (sensitivity). Compare with re-solving the LP.

(6) If each order increases by +5%, estimate the variation of optimal cost via dual variables and compare with the true value obtained by modifying the data file.

Notes: This exercise sheet is taken from the course material (practical optimization Lesson 5). For data files and further instructions (AMPL files, `cheb.dat`, etc.) refer to the course platform. :contentReference[oaicite:1]index=1